

Supplementary theory and analyses for “Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China”

Jian Hou¹, Tan Meng² and Maozai Tian^{2,*}

¹College of Systems Engineering, National University of Defense Technology, Changsha 410073, China

²Center for Applied Statistics, School of Statistics, Renmin University of China, 59 Zhongguancun Street, Beijing 100872, China

*Corresponding author. mztian@ruc.edu.cn

This supplement provides the proof of Proposition 1, a process-level strong-mixing result, additional applied analyses and the simulation specifications and results. Unless stated otherwise, evaluation-period tables use 33 summers and exclude the 2015 and 2022 method-development summers. The historical and station extensions state their own temporal supports.

Analysis sequence

Table S1 summarises the sequence of analyses. Unless stated otherwise, each calculation uses the 121-site labels, five bandwidths and equal-month, equal-scale and equal-summer estimator. The historical and station extensions were designed before their respective data were retrieved.

Table S1: Sequence and purpose of the applied analyses.

Element	Analytical status	Data or support	Scientific purpose
Field construction, event labels, five graph scales and aggregation	Primary specification	2015 and 2022 design summers	Definition of the evaluation-period estimand
33-summer evaluation-period contrast	Primary finite-record estimate	33 evaluation summers	Evaluation-period estimate
Global 99,999-draw product shift	Exploratory inferential analysis	33 evaluation summers	Conditional randomisation under the product-invariance null
Process-model references, alternative effect measures, spatial bases, bandwidth and support comparisons	Exploratory analyses	33 evaluation summers and outcome-free simulations	Reference inference, structural decomposition and sensitivity analyses
Historical climatology–anomaly decomposition	Exploratory structural analysis	41 historical summers	Period-specific accounting of anomaly energy and climatology–anomaly alignment
1950–1990 ERA5-Land extension	Pre-access extension	41 JJA summers	Temporal extension and its product-shift calculation

Continued on the next page

Table S1 continued

Element	Analytical status	Data or support	Scientific purpose
Ten-year NOAA-ERA5-Land extension	Pre-access extension	Ten evaluation years	External measurement and temporally separated effect comparison
Heavy-tail, coupled-field and cross-record simulations	Exploratory simulations	Outcome-free simulated records	Estimator behaviour and randomisation calibration

S1. Finite-record product cyclic-shift validity

Let X collect the regional-mean records used to define the event labels, and let D collect the corresponding daily graph-dispersion profiles. The finite group G acts on D by independently rotating the complete multiscale profile within each month-year record. Conditional on X , suppose

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \quad (\text{S1})$$

For a measurable lower-tail statistic $T(X, D)$ fixed before the transformations are examined, define

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \leq T(X, D)\}.$$

Proposition 1 (Product cyclic-shift validity) *For every $u \in [0, 1]$,*

$$\Pr\{p_G(X, D) \leq u \mid X\} \leq u \quad \text{almost surely.}$$

If the $|G|$ transformed statistics have no ties, p_G is uniform on $\{1/|G|, \dots, 1\}$ conditional on the invariant orbit. If $H_1, \dots, H_B$ are independent uniforms on G , independently of (X, D) , then

$$p_B = \frac{1 + \sum_{b=1}^B I\{T(X, H_b D) \leq T(X, D)\}}{B + 1}$$

satisfies

$$\Pr\{p_B \leq u \mid X\} \leq \frac{\lfloor (B+1)u \rfloor}{B+1} \leq u.$$

For fixed scale statistics $T_1, \dots, T_H$, put $M(X, D) = \min_h T_h(X, D)$ and define

$$p_h^{\text{adj}} = \frac{1}{|G|} \sum_{g \in G} I\{M(X, gD) \leq T_h(X, D)\}.$$

Under the complete joint invariance null, $\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u$. This is weak family-wise control; the proposition does not give strong control under arbitrary partial nulls.

Proof Fix a value x in the probability-one set on which the conditional invariance in Equation (S1) holds, and write $S(y) = T(x, y)$. Let $m = |G|$.

For fixed y , form the multiset $\{S(gy) : g \in G\}$. For each $g \in G$, let

$$r_g(y) = \sum_{h \in G} I\{S(hy) \leq S(gy)\}.$$

Right multiplication by g permutes the elements of G . Hence

$$\begin{aligned} p_G(x, gy) &= \frac{1}{m} \sum_{h \in G} I\{S(hgy) \leq S(gy)\} \\ &= \frac{r_g(y)}{m}. \end{aligned}$$

If U is uniform on G , independently of y , then

$$\Pr\{p_G(x, Uy) \leq u \mid y\} = \frac{1}{m} \#\{g : r_g(y) \leq mu\}. \quad (\text{S2})$$

Order the m values in the multiset. An observation at the upper end of a tie block has weak rank equal to the final position of that block. Every tie block counted on the right of Equation (S2) therefore ends no later than position $\lfloor mu \rfloor$. The number of counted group elements is at most $\lfloor mu \rfloor$, and

$$\Pr\{p_G(x, Uy) \leq u \mid y\} \leq \frac{\lfloor mu \rfloor}{m} \leq u.$$

This inequality also holds when the action has a nontrivial stabiliser. Repeated transformations then appear as ties in the multiset and make the weak-rank test conservative.

Let U now be independent of (X, D) and uniform on G . Conditional invariance gives $UD \mid X = x \stackrel{d}{=} D \mid X = x$. Applying the preceding inequality conditionally on D yields

$$\begin{aligned} \Pr\{p_G(x, D) \leq u \mid X = x\} &= \Pr\{p_G(x, UD) \leq u \mid X = x\} \\ &= \mathbb{E}[\Pr\{p_G(x, UD) \leq u \mid X = x, D\} \mid X = x] \\ &\leq u. \end{aligned}$$

If the transformed statistics have no ties, the ranks $r_g(y)$ are $1, \dots, m$ once each. Equality holds at $u = j/m$, proving uniformity on the attainable grid.

For the sampled version, take $U_0, U_1, \dots, U_B$ independently and uniformly from G , independently of (X, D) . Conditional on (X, D) , the variables $S(U_0D), \dots, S(U_BD)$ are identically distributed and exchangeable. The weak rank of the first among these $B + 1$ values satisfies

$$\Pr \left[\frac{1 + \sum_{b=1}^B I\{S(U_bD) \leq S(U_0D)\}}{B + 1} \leq u \mid X = x \right] \leq \frac{\lfloor (B + 1)u \rfloor}{B + 1}. \quad (\text{S3})$$

It remains to relate the random anchor to the implemented construction, whose first transformation is the identity. Put $H_b = U_b U_0^{-1}$ for $b = 1, \dots, B$ and $D^* = U_0 D$. The map

$$(U_0, U_1, \dots, U_B) \mapsto (U_0, H_1, \dots, H_B)$$

is a bijection on G^{B+1} . Thus $H_1, \dots, H_B$ are independent uniforms and are independent of U_0 . Moreover, $D^* \mid X = x \stackrel{d}{=} D \mid X = x$. The vector in Equation (S3) consequently has the same conditional distribution as

$$\{S(D), S(H_1D), \dots, S(H_BD)\}.$$

This is the identity-plus- B construction used to calculate p_B .

The statistic must remain a fixed measurable function of the jointly shifted outcome matrix. For the scale-wise single-step comparison, let T_h denote the scale statistics and put $M(y) = \min_h T_h(x, y)$. The adjusted value

$$p_h^{\text{adj}} = \frac{1}{m} \sum_{g \in G} I\{M(gy) \leq T_h(x, y)\}$$

is nondecreasing in its observed threshold. Therefore $\min_h p_h^{\text{adj}}$ is attained at a scale for which $T_h(x, y) = M(y)$ and equals the randomisation value for M . Applying the enumeration result to M gives

$$\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u.$$

The conclusion uses the complete joint null because M combines every scale. This completes the proof. $\square$

S2. Process-level strong-mixing and HAC reference

Let $\hat{\mathbf{R}}_r = (\hat{R}_{h_1 r}, \dots, \hat{R}_{h_H r})^\top$ be the scale profile for summer r , $\mathbf{a} = H^{-1} \mathbf{1}_H$, $Z_r = \mathbf{a}^\top \hat{\mathbf{R}}_r$ and $\hat{\theta}_R = R^{-1} \sum_{r=1}^R Z_r$. Write $\gamma(k) = \text{Cov}(Z_0, Z_k)$ and define

$$\begin{aligned} \hat{\gamma}_R(k) &= \frac{1}{R} \sum_{r=k+1}^R (Z_r - \bar{Z}_R)(Z_{r-k} - \bar{Z}_R), \\ \hat{\sigma}_R^2 &= \hat{\gamma}_R(0) + 2 \sum_{k=1}^{L_R} K\left(\frac{k}{L_R + 1}\right) \hat{\gamma}_R(k). \end{aligned} \quad (\text{S4})$$

Theorem S1 (Strong-mixing limit and HAC reference) *Suppose $\{\hat{\mathbf{R}}_r\}_{r \in \mathbb{Z}}$ is strictly stationary and strongly mixing, with $\mathbb{E}\|\hat{\mathbf{R}}_0\|^{2+\delta} < \infty$ for some $\delta > 0$ and $\sum_{k=1}^{\infty} \alpha(k)^{\delta/(2+\delta)} < \infty$. Then every entry of*

$$\boldsymbol{\Omega} = \sum_{k=-\infty}^{\infty} \text{Cov}(\hat{\mathbf{R}}_0, \hat{\mathbf{R}}_k)$$

is absolutely convergent. If $\sigma^2 = \mathbf{a}^\top \boldsymbol{\Omega} \mathbf{a} > 0$ and $\theta = \mathbf{a}^\top \mathbb{E}(\hat{\mathbf{R}}_0)$, then

$$\sqrt{R}(\hat{\theta}_R - \theta) \xrightarrow{d} N(0, \sigma^2).$$

Suppose in addition that K is bounded on $[0, \infty)$, $K(0) = 1$, is continuous at zero and vanishes outside $[0, 1]$; $L_R \rightarrow \infty$, $L_R/R \rightarrow 0$; and

$$\sum_{k=0}^{L_R} |\hat{\gamma}_R(k) - \gamma(k)| = o_p(1). \quad (\text{S5})$$

Then $\hat{\sigma}_R^2 \xrightarrow{p} \sigma^2$ and

$$\frac{\sqrt{R}(\hat{\theta}_R - \theta)}{\hat{\sigma}_R} \xrightarrow{d} N(0, 1).$$

For a ratio component $\hat{R}_{jr} = A_{jr}/B_{jr} - 1$, with $A_{jr} \geq 0$ and $B_{jr} > 0$, a sufficient condition for its $(2 + \delta)$ moment is the existence of conjugate $p, q > 1$ such that

$$\mathbb{E} A_{j0}^{(2+\delta)p} < \infty, \quad \mathbb{E} B_{j0}^{-(2+\delta)q} < \infty.$$

The same conclusion holds for a finite average of such ratios when the condition holds for every constituent ratio.

Proof Let $\boldsymbol{\mu} = E(\widehat{\mathbf{R}}_0)$. A measurable transformation cannot increase strong-mixing coefficients, so $\{Z_r\}$ is strictly stationary and strongly mixing with coefficients bounded by $\alpha(k)$. The inequality $|\mathbf{a}^\top v| \leq \|\mathbf{a}\| \|v\|$ gives $E|Z_0|^{2+\delta} < \infty$.

Put $\xi_r = Z_r - E Z_0$. Davydov's covariance inequality, with both moment exponents equal to $2 + \delta$, gives a finite constant C such that

$$|\gamma(k)| = |E(\xi_0 \xi_k)| \leq C \|\xi_0\|_{2+\delta}^2 \alpha(|k|)^{\delta/(2+\delta)}, \quad k \neq 0. \quad (\text{S6})$$

The mixing-rate assumption implies $\sum_{k=-\infty}^{\infty} |\gamma(k)| < \infty$. Applying the same inequality componentwise establishes absolute convergence of every entry of $\boldsymbol{\Omega}$. Bilinearity and absolute convergence permit interchange of the finite quadratic form and the infinite sum:

$$\mathbf{a}^\top \boldsymbol{\Omega} \mathbf{a} = \sum_{k=-\infty}^{\infty} \text{Cov}(Z_0, Z_k) = \sum_{k=-\infty}^{\infty} \gamma(k) = \sigma^2. \quad (\text{S7})$$

The central limit theorem for strongly mixing sequences applies under the stated moment and mixing-rate conditions (Ibragimov and Linnik, 1971). Hence

$$R^{-1/2} \sum_{r=1}^R \xi_r \xrightarrow{d} N(0, \sigma^2).$$

Since $E Z_0 = \mathbf{a}^\top \boldsymbol{\mu} = \theta$ and $R^{-1} \sum_r Z_r = \widehat{\theta}_R$, this proves the first limit.

For consistency of Equation (S4), define

$$\sigma_{K, L_R}^2 = \gamma(0) + 2 \sum_{k=1}^{L_R} K \left(\frac{k}{L_R + 1} \right) \gamma(k).$$

Boundedness of K and Equation (S5) give

$$\begin{aligned} |\widehat{\sigma}_R^2 - \sigma_{K, L_R}^2| &\leq |\widehat{\gamma}_R(0) - \gamma(0)| \\ &\quad + 2 \|K\|_\infty \sum_{k=1}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1). \end{aligned}$$

For each fixed k , continuity at zero and $L_R \rightarrow \infty$ imply $K\{k/(L_R + 1)\} \rightarrow 1$. The weights are uniformly bounded, vanish for $k > L_R$, and $\sum_k |\gamma(k)| < \infty$. Dominated convergence for series therefore yields

$$\sigma_{K, L_R}^2 \longrightarrow \gamma(0) + 2 \sum_{k=1}^{\infty} \gamma(k) = \sigma^2.$$

Thus $\widehat{\sigma}_R^2 \xrightarrow{p} \sigma^2$. Positivity of σ^2 and Slutsky's theorem give the studentised limit.

Condition (S5) is separate from the assumptions used for the central limit theorem. Primitive sufficient conditions for it require higher moments, stronger mixing-rate summability and a compatible bandwidth rate (Andrews, 1991; Newey and West, 1987).

For comparison, fix a lag L . Strong mixing implies ergodicity. The ergodic theorem applied to $(Z_r - E Z_0)(Z_{r-k} - E Z_0)$, together with $\overline{Z}_R \xrightarrow{p} E Z_0$, gives $\widehat{\gamma}_R(k) \xrightarrow{p} \gamma(k)$ for each fixed k . The fixed-lag estimator therefore converges to

$$\sigma_{K, L}^2 = \gamma(0) + 2 \sum_{k=1}^L K \left(\frac{k}{L+1} \right) \gamma(k). \quad (\text{S8})$$

For Bartlett lag $L = 2$, this is $\gamma(0) + 2\{(2/3)\gamma(1) + (1/3)\gamma(2)\}$. Its difference from the full long-run variance is

$$\sigma^2 - \sigma_{K, 2}^2 = 2 \left\{ \frac{1}{3} \gamma(1) + \frac{2}{3} \gamma(2) + \sum_{k=3}^{\infty} \gamma(k) \right\}.$$

The fixed-lag estimator is consistent only if this weighted error is zero.

It remains to verify the stated sufficient ratio condition. With $s = 2 + \delta$, convexity and Hölder's inequality give

$$\begin{aligned} \mathbb{E} |A_{j0}/B_{j0} - 1|^s &\leq 2^{s-1} \left\{ \mathbb{E}(A_{j0}^s B_{j0}^{-s}) + 1 \right\} \\ &\leq 2^{s-1} \left[\{\mathbb{E} A_{j0}^{sp}\}^{1/p} \{\mathbb{E} B_{j0}^{-sq}\}^{1/q} + 1 \right] < \infty. \end{aligned}$$

A finite sum of variables with finite s th moments also has a finite s th moment. Applying this argument to each component proves the final claim. $\square$

The 495 observed middle-day denominators range from 2.25 to 25.99 $^{\circ}\text{C}^2$. The process-level ratio argument additionally assumes the negative moment stated in Theorem S1.

S3. Global product-shift result

The product-shift analysis used the statistic $T = \hat{\theta}_{y_H}$. Each draw independently selected one cyclic offset in each of the 99 evaluation month-year records and shifted all five graph-dispersion columns together. Regional-mean series and event labels remained in their observed order. The statistic was then reconstructed with equal months, scales and summers. The generator seed was 20260809 and $B = 99,999$.

No sampled statistic was at or below the observed -7.2834% . The plus-one value is therefore $1/(99,999 + 1) = 0.00001$, its minimum attainable value for this run. The sampled null had mean 0.8241% , standard deviation 1.5651 points and median 0.8139% ; its 0.1% , 1% and 5% quantiles were -3.8788% , -2.7440% and -1.7225% . A ratio-based randomisation distribution need not be centred at zero, and the test compares the observed statistic with that distribution directly.

Table S2. The 99,999-draw product cyclic-shift analysis. Effects, null means and null standard deviations are in percentage points. The last column uses the minimum statistic over the five scales and has weak family-wise validity only under the complete joint invariance null.

Bandwidth (km)	Observed (points)	Null mean (points)	Null SD (points)	Unadjusted p	Complete-null p
126	-2.856	0.359	1.059	0.00104	0.02504
252	-3.945	0.531	1.301	0.00022	0.00571
503	-5.907	0.859	1.683	0.00002	0.00019
1,006	-10.441	1.125	1.964	0.00001	0.00001
2,013	-13.268	1.246	2.089	0.00001	0.00001
Five-scale mean	-7.283	0.824	1.565	0.00001	—

The relevant null is the conditional product-invariance statement in Equation (S1). The scale-wise rows use the same joint draws and are mutually dependent.

S4. Physical scale of the graph contrast

For a field y ,

$$Q_h(y) = \frac{\sum_{i < j} w_{ij,h} (y_i - y_j)^2}{2 \sum_{i < j} w_{ij,h}}, \quad D_h^{\text{RMS}}(y) = \{2Q_h(y)\}^{1/2}.$$

Thus Q_h has units $^{\circ}\text{C}^2$ and D_h^{RMS} is the weighted root mean squared pairwise difference in $^{\circ}\text{C}$. Table S3 reports equal-month and equal-summer means of Q_h and applies $D_h^{\text{RMS}} = (2Q_h)^{1/2}$ to each displayed mean. The relative columns are means of record-level ratios, so they are not obtained by taking ratios of the displayed marginal means.

The direct five-scale mean of the record-level RMS changes is -3.97% . Applying the square-root transformation only after averaging the five graph-dispersion effects gives $\sqrt{1 - 0.072834} - 1 = -3.71\%$. The two values differ because

Table S3. Physical scale of graph dispersion in the evaluation summers. The RMS columns are $(2Q_h)^{1/2}$ transformations of the displayed mean Q_h columns. The interval is a 95% t -based interval that treats summers as independent. RMS change is the mean record-level relative change in D_h^{RMS} .

Bandwidth (km)	Q_h^M (°C ^{1/2})	Q_h^H (°C ^{1/2})	RMS ^M (°C)	RMS ^H (°C)	Q_h effect [interval] (%)	RMS change (%)
126	2.662	2.578	2.307	2.271	−2.86 [−5.04, −0.67]	−1.54
252	4.356	4.169	2.952	2.887	−3.94 [−6.50, −1.39]	−2.14
503	7.943	7.433	3.986	3.856	−5.91 [−8.91, −2.90]	−3.25
1,006	12.892	11.489	5.078	4.793	−10.44 [−13.49, −7.39]	−5.70
2,013	15.903	13.743	5.640	5.243	−13.27 [−16.28, −10.26]	−7.22

the square root and the averaging steps do not commute. In either form, a percentage change in squared pairwise differences is larger in magnitude than the corresponding change in RMS amplitude.

S5. Structural decompositions

S5.1. Climatology–anomaly energy

Let μ_m be the site-specific 35-summer mean field for month m and write $\mathbf{y}_t = \mu_m + \mathbf{a}_t$. For a fixed graph,

$$Q_h(\mathbf{y}_t) = \frac{\mu_m^\top \mathbf{L}_h \mu_m}{2S_h} + \frac{\mu_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \quad (\text{S9})$$

The first term is fixed within month m , so its high-minus-middle difference is exactly zero. Within each record, we divide the differences of the other two terms by the middle-day total Q_h mean. The resulting anomaly-energy and climatology–anomaly alignment components add to the original relative contrast before any aggregation.

Table S4. Exact climatology–anomaly decomposition. Entries are percentage points; brackets are 95% intervals. The climatology–energy difference is zero by construction.

Bandwidth (km)	Total effect [interval]	Anomaly energy [interval]	Climatology–anomaly alignment [interval]
126	−2.86 [−5.04, −0.67]	−2.63 [−3.55, −1.72]	−0.22 [−1.98, 1.54]
252	−3.94 [−6.50, −1.39]	−2.45 [−3.45, −1.45]	−1.49 [−3.69, 0.70]
503	−5.91 [−8.91, −2.90]	−1.42 [−2.43, −0.41]	−4.49 [−7.24, −1.74]
1,006	−10.44 [−13.49, −7.39]	−0.18 [−1.10, 0.74]	−10.26 [−13.20, −7.32]
2,013	−13.27 [−16.28, −10.26]	+0.41 [−0.50, 1.31]	−13.68 [−16.68, −10.67]

At 2,013 km, anomaly energy is lower in 16 of 33 summers, whereas the alignment component is negative in all 33. The corresponding counts at 1,006 km are 21 and 28. The broad-scale contraction is therefore associated with daily anomalies opposing the persistent monthly field, while the transient anomaly energy itself changes little. The decomposition is descriptive and conditional on the chosen climatology. The maximum aggregated identity error over the five graphs is 1.82×10^{-16} .

S5.2. Geographic and topographic bases

For a centred basis matrix $\mathbf{B}$, define the graph-orthogonal projection

$$\hat{\beta}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{L}_h \mathbf{y}_t, \quad \hat{\mathbf{y}}_{th} = \mathbf{B} \hat{\beta}_{th}.$$

The structured energy of $\hat{\mathbf{y}}_{th}$ and the residual energy of $\mathbf{y}_t - \hat{\mathbf{y}}_{th}$ add exactly to $Q_h(\mathbf{y}_t)$. Table S5 compares centred latitude, the latitude–longitude plane and a latitude–longitude–elevation basis. Elevation comes from the invariant ERA5-Land geopotential field: the nearest 0.1-degree grid value at each analysis site was divided by 9.80665 m s^{-2} .

The 121 elevations range from -0.9 to $1,940.3$ m. Each basis column was centred and scaled before the projection. Such scaling leaves its column space unchanged.

Table S5. Nested spatial-basis comparison. Values are mean percentage-point contributions across the evaluation summers. The structured and residual columns for each basis add to the total, apart from rounding.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Geographic– topographic structured	Geographic– topographic residual
126	−2.86	−2.58	−0.28	−2.81	−0.04	−1.16	−1.69
252	−3.94	−4.60	+0.66	−5.07	+1.13	−2.85	−1.10
503	−5.91	−8.38	+2.48	−9.32	+3.42	−6.13	+0.23
1,006	−10.44	−12.73	+2.29	−13.94	+3.50	−11.39	+0.94
2,013	−13.27	−14.96	+1.69	−16.23	+2.96	−14.34	+1.07

The latitude component is negative in 33 of 33 summers at every scale. The planar component is negative in 32 summers at 126 km and all 33 summers at the other four scales. At 2,013 km, its 95% interval is $[-18.16, -14.30]$ points and the residual interval is $[1.59, 4.33]$ points. Adding longitude makes the structured component more negative and the compensating residual more positive.

The geographic–topographic component is negative in 18, 21, 28, 33 and 33 of the 33 summers as bandwidth increases. At 126 km its estimate is -1.16 points with a 95% interval of $[-2.66, 0.33]$, which includes zero. At 2,013 km the estimate is -14.34 points $[-16.75, -11.92]$, and the residual is $+1.07$ points. The broad contraction persists after adding elevation to the latitude–longitude basis. The projection yields a combined geographic–topographic contribution; no order-invariant elevation contribution or physical mechanism follows from it. The maximum day-level identity error for the augmented basis is 8.88×10^{-14} , and the maximum aggregated error is 2.78×10^{-16} .

S6. Scale and support resolution

S6.1. Thirty-one-point bandwidth curve

The estimand is the equal-weight mean over five design bandwidths. A denser curve evaluates 31 logarithmically spaced values between 125.800 and 2,012.796 km on the same 121 sites, using the evaluation fields and labels. The five design bandwidths define the estimand; the denser curve describes its scale profile.

Table S6. The 31-point bandwidth curve. Values are mean graph-dispersion effects in per cent. The three column pairs continue the same ordered curve.

h (km)	Effect	h (km)	Effect	h (km)	Effect
125.8	−2.856	347.7	−4.596	876.1	−9.538
138.0	−2.958	381.4	−4.835	961.0	−10.151
151.3	−3.076	418.3	−5.125	1,054.0	−10.720
166.0	−3.211	458.8	−5.479	1,156.0	−11.239
182.1	−3.360	503.2	−5.907	1,268.0	−11.703
199.7	−3.521	551.9	−6.407	1,390.8	−12.113
219.0	−3.688	605.4	−6.974	1,525.4	−12.470
240.2	−3.859	664.0	−7.591	1,673.1	−12.778
263.5	−4.031	728.3	−8.239	1,835.1	−13.043
289.0	−4.206	798.8	−8.895	2,012.8	−13.268
317.0	−4.391				

All 30 adjacent changes are negative, and all 31 pointwise 95% intervals have upper endpoints below zero. The five design estimates remain $-2.86, -3.94, -5.91, -10.44$ and -13.27% . The curve shows no reversal at the 31 evaluated bandwidths. The intervals are descriptive pointwise summaries; no simultaneous confidence band is estimated.

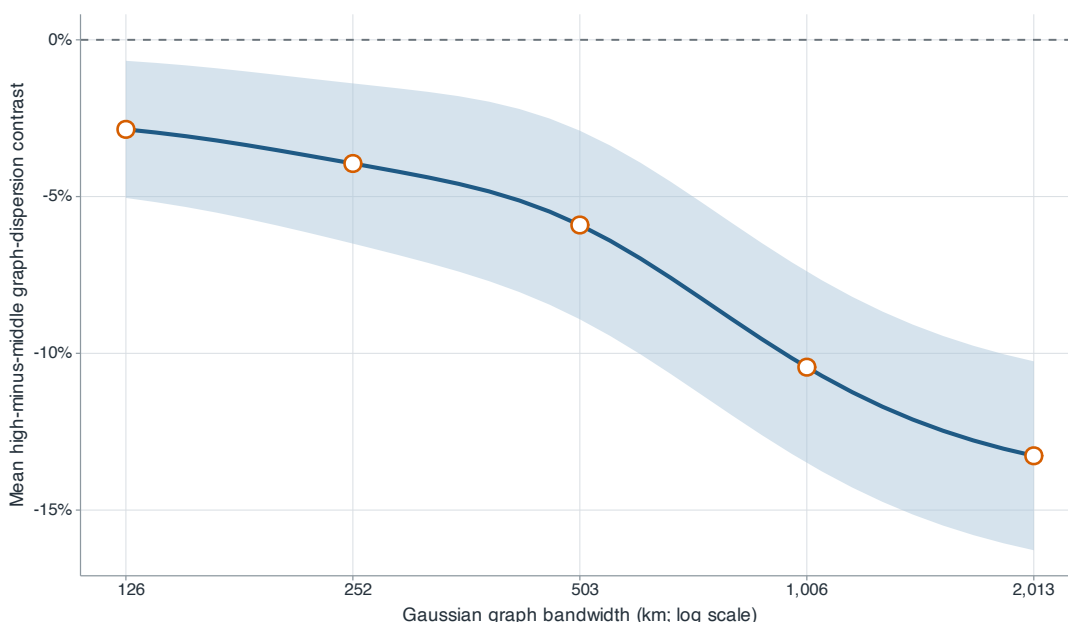


Figure 1 Bandwidth-resolution profile on the 121-site support. The blue line and shaded band show the 31-point profile and its pointwise 95% intervals; orange open circles mark the five design bandwidths.

S6.2. Four spatial-support configurations

The convergence comparison uses the five physical bandwidths, evaluation peak times and 121-site labels. The 239-site configuration refines longitude; the 236-site configuration refines latitude. Each contains all 121 sites and is a subset of the 465-site grid. The two one-direction refinements differ in their added sites.

Table S7. Four-configuration spatial-resolution comparison. Effects and the two error summaries are in percentage points. Differences use the 465-site result at the same physical bandwidth as the reference.

Configuration	Sites	126	252	503	1,006	2,013	Max. $ \Delta $	RMSE
Analysis support	121	-2.856	-3.945	-5.907	-10.441	-13.268	0.991	0.764
Longitude refined	239	-2.839	-3.846	-5.968	-10.552	-13.351	0.929	0.728
Latitude refined	236	-3.129	-4.308	-6.529	-10.865	-13.458	0.500	0.384
Dense support	465	-3.395	-4.543	-6.898	-11.340	-13.959	0.000	0.000

Every support retains the broad-to-local ordering and a negative effect at all five bandwidths. Relative to 465 sites, the 121-site profile has a maximum absolute discrepancy of 0.99 points and an across-scale RMSE of 0.76 points. These finite-grid comparisons show how the estimates change with spatial support. Agreement is weakest for the narrowest graph, consistent with its shorter length scale relative to intersite spacing.

S7. Historical extension, 1950–1990

The ERA5-Land archive distributed through the Copernicus Climate Data Store begins in January 1950.¹ A plan dated 9 August 2026 defined the 41 summers, 121 sites, field construction, five scales, aggregation and product-shift calculation before the historical values were retrieved. The analysis weights the three months, five scales and summers equally. The two periods are separated in calendar time but share the ERA5-Land model and assimilation system.

¹ Dataset overview: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview>.

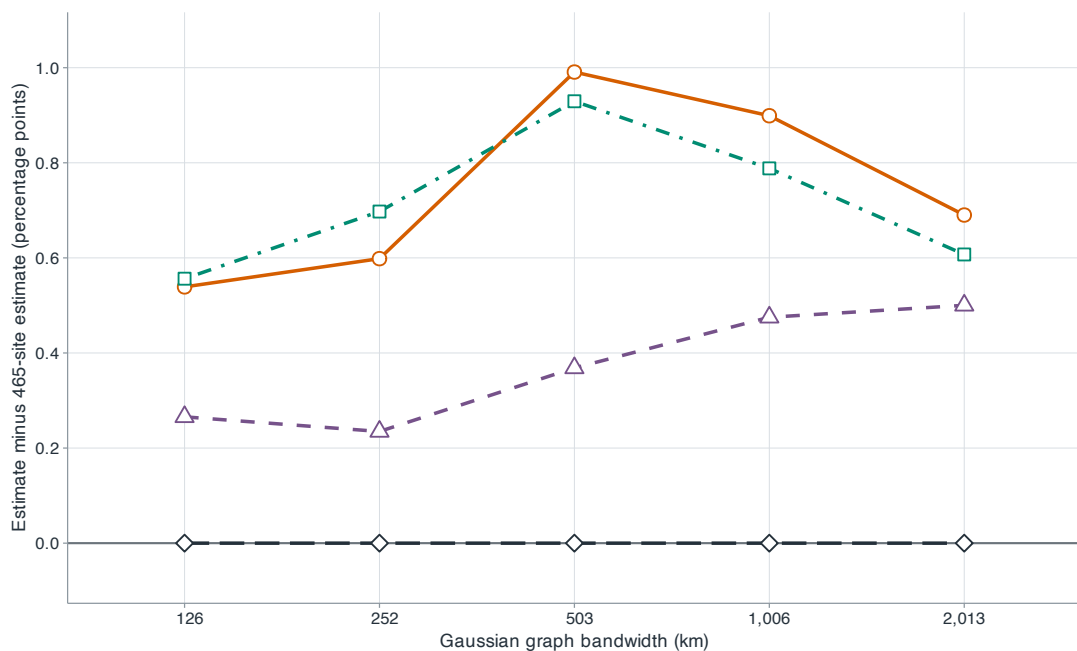


Figure 2 Differences from the 465-site estimate at five bandwidths. Orange circles with a solid line denote the 121-site support; purple triangles with a dashed line, the latitude-refined 236-site support; teal squares with a dot-dash line, the longitude-refined 239-site support; and dark diamonds with a long-dash line, the dense 465-site reference.

The product-shift calculation uses 99,999 draws, seed 20260810 and joint rotation of all five dispersion columns within each month-year record.

The five-scale effect over 1950–1990 is -11.04% (95% interval $[-12.84, -9.24]\%$), with a negative estimate in 40 of 41 summers. Deleting one summer at a time gives estimates from -11.33% to -10.62% . Table S8 gives the period split and the five-scale profile.

Table S8. The 1950–1990 historical extension. Estimates and intervals are percentages; the final column counts summers with a negative estimate. Period rows average the five bandwidths. Scale rows use all 41 summers.

Period or bandwidth	Summers	Estimate	95% interval	Negative summers
1950–1990 overall	41	−11.04	[−12.84, −9.24]	40
1950–1978	29	−10.50	[−12.50, −8.51]	28
1979–1990	12	−12.34	[−16.65, −8.04]	12
126 km	41	−5.96	[−7.37, −4.54]	38
252 km	41	−7.51	[−9.15, −5.87]	39
503 km	41	−10.08	[−12.11, −8.04]	40
1,006 km	41	−14.50	[−16.72, −12.27]	41
2,013 km	41	−17.17	[−19.44, −14.91]	41

Provider documentation identifies the forcing change used for the period split: ERA5-Land for 1950–1978 used the preliminary ERA5 back extension, whereas the later period used the consolidated ERA5 release.² Both period estimates are negative, and their intervals overlap. The 1950–1978 estimate inherits uncertainty from the preliminary forcing data.

² ERA5-Land production note: <https://confluence.ecmwf.int/pages/viewpage.action?pageId=402639006>.

We applied the product cyclic-shift calculation to 123 month-year records. None of the sampled statistics was at or below the observed -11.043% , so the plus-one lower-tail value is 0.00001. The sampled null mean and standard deviation are 0.853% and 1.436 percentage points. Every scale has an unadjusted value of 0.00001; the complete-null single-step values range from 0.00001 to 0.00006.

For the historical climatology–anomaly decomposition, the monthly reference field was estimated within 1950–1990:

$$\mu_m^H = \frac{1}{41} \sum_{y=1950}^{1990} \left(\frac{1}{n_m} \sum_{t \in m} y_{yt} \right), \quad \mathbf{a}_{yt}^H = y_{yt} - \mu_m^H.$$

Thus μ_m^H is the site-specific mean field for month m across the 41 historical summers. The anomaly and alignment terms therefore describe departures from the historical period's own persistent monthly spatial pattern.

Table S9 reports the latitude, planar and climatology–anomaly accounting for the historical extension. The latitude and planar structured components are negative in all 41 summers at every bandwidth. At 2,013 km, anomaly energy is -0.28 points with interval $[-1.01, 0.45]$, while the climatology–anomaly alignment component is -16.89 points $[-19.31, -14.47]$ and is negative in all 41 summers. The broad-scale historical result is therefore also alignment-dominated, and its latitude and planar components have the same negative direction as in the 1991–2025 period.

Table S9. Structural accounting for the 1950–1990 extension. Entries are percentage points. Within each geographic basis, the structured and residual columns add to the total. The anomaly and alignment columns also add to the total because the climatology–energy difference is zero.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Anomaly energy	Climatology– anomaly alignment
126	−5.96	−2.83	−3.13	−3.15	−2.81	−3.22	−2.74
252	−7.51	−4.92	−2.59	−5.50	−2.01	−3.30	−4.21
503	−10.08	−8.86	−1.21	−9.92	−0.15	−2.41	−7.67
1,006	−14.50	−13.60	−0.90	−14.99	+0.49	−1.02	−13.48
2,013	−17.17	−16.13	−1.05	−17.62	+0.44	−0.28	−16.89

S8. NOAA station extension in ten evaluation years

The station comparison uses the ten summers enumerated before station retrieval: 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025. The set is $\{1992 + 4k : k = 0, \dots, 7\} \cup \{2023, 2025\}$. The official Integrated Surface Database (ISD) station-history snapshot used for selection was archived.³ The planned eligibility rule required a station-history end date on or after 31 August 2025, but that snapshot contained no such candidate: its global maximum end date was 28 August and its maximum inside the study rectangle was 24 August. The operational administrative cutoff was therefore 24 August. This administrative change preceded retrieval of station WBT observations and matched ERA5-Land values.

For a candidate station s , the proximity criterion was

$$d(s, \mathcal{S}_{121}) = \min_{i \in \mathcal{S}_{121}} d(s, i) \leq 150 \text{ km},$$

where $\mathcal{S}_{121}$ is the 121-site analysis support. For coordinate differences in degrees, the pairwise distance was

$$d(s, i) = \left[\{111.32 \cos(\bar{\phi}_{si})(\lambda_s - \lambda_i)\}^2 + \{110.57(\phi_s - \phi_i)\}^2 \right]^{1/2},$$

with $\bar{\phi}_{si}$, the pair's mean latitude, converted to radians in the cosine. This filter was applied before maximin selection and restricted station locations to within 150 km of the analysis support. Of the 228 stations satisfying the

³ NOAA Integrated Surface Database: <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>.

rectangle and date conditions, 20 exceeded 150 km, leaving 208 candidates; the largest retained nearest-site distance was 147.73 km and the smallest excluded distance was 155.72 km. A deterministic maximin rule based on station coordinates selected 30 stations. The study rectangle defined the candidate domain. A station-year qualified with at least 400 exact UTC hours having temperature, dew point and usable station pressure; 26–30 of the 30 selected stations qualified in each year. Station wet-bulb temperature was calculated with the same Bolton implementation used for ERA5-Land.

The nearest ERA5-Land cells for the Shengsi and Taidong island stations had no finite land values at any requested time, so both point series were recorded as unavailable. The other 28 station locations had finite ERA5-Land series. Combining this land-mask restriction with the hourly qualification rule left 24–28 matched stations per year and 175,172 paired hours. Table S10 reports the measurement comparison. Bias is ERA5-Land minus station wet-bulb temperature; the centred correlation subtracts each station mean before pooling. The all-year row pools all matched hours across years.

Table S10. ERA5-Land and NOAA station wet-bulb agreement in ten evaluation years. Bias, MAE and RMSE are in °C. Stations are the qualified locations with a finite nearest-cell ERA5-Land series in that year.

Year	Stations	Matched hours	Bias	MAE	RMSE	Centred correlation
1992	25	17,887	−0.25	0.88	1.16	0.919
1996	25	19,147	−0.33	0.88	1.16	0.908
2000	25	16,765	−0.40	0.94	1.21	0.908
2004	25	16,829	−0.20	0.91	1.18	0.923
2008	25	17,310	−0.20	0.88	1.14	0.909
2012	25	17,349	−0.27	0.93	1.22	0.918
2016	24	16,589	−0.34	0.88	1.13	0.937
2020	28	19,323	−0.26	1.04	1.48	0.897
2023	27	16,554	−0.19	1.02	1.49	0.925
2025	28	17,419	−0.17	1.01	1.45	0.933
All years	28	175,172	−0.26	0.94	1.27	0.916

For the effect comparison, ERA5-Land peak times and high/middle labels defined the event days, and we aligned station observations to those events. Each graph field used the same available station subset for NOAA and ERA5-Land and required at least ten stations. All 30 month-year records retained at least one high and one middle field, giving 345 usable event fields. We formed high-to-middle ratios within records, then averaged them equally over months within years and over the ten years. Table S11 reports 95% intervals. Inference focuses on the 503-, 1,006- and 2,013-km graphs supported by the station spacing; the two narrower graphs remain descriptive.

Table S11. High–middle graph-dispersion effects at ERA5-Land-defined peak times and labels in ten evaluation years. Effects and intervals are percentages. “Negative” counts years with a negative scale-specific estimate. Each scale contains all 30 month-year records and an average of 25.3 stations per event field.

Bandwidth (km)	NOAA effect [95% interval]	NOAA negative years	ERA5-Land effect [95% interval]	ERA negative years
126	−6.23 [−25.87, 13.41]	6/10	−11.60 [−27.49, 4.29]	7/10
252	−7.23 [−21.01, 6.55]	7/10	−12.74 [−22.68, −2.81]	8/10
503	−12.50 [−24.05, −0.96]	8/10	−17.13 [−25.15, −9.11]	10/10
1,006	−18.03 [−29.17, −6.90]	9/10	−21.60 [−29.07, −14.14]	10/10
2,013	−21.47 [−32.51, −10.43]	9/10	−24.23 [−31.72, −16.74]	10/10

The station and paired ERA5-Land profiles are negative at all five scales and become more negative with bandwidth. Averaging the three station-supported scales within each year gives −17.34% (95% interval [−28.49, −6.18]%), with a negative average in nine of ten years. The same broad-profile calculation for the matched ERA5-Land fields gives −20.99% ([−28.57, −13.40]%), negative in all ten years. The two fields are compared on identical, time-varying station supports. Because ERA5-Land defines the event times and labels, the comparison measures agreement for common events on a sparse station network. Coverage is weakest at 126 and 252 km.

S9. Continuous intensity and calendar matching

Two alternative event contrasts use the same sites, graphs and aggregation hierarchy. The continuous calculation regresses $\log Q_h$ on centred regional wet-bulb temperature separately in each month-year record, then averages the slopes equally over months, scales and the 33 evaluation summers. The five-scale mean slope is $-0.0650\text{ }^{\circ}\text{C}^{-1}$, with a 95% interval of $[-0.0822, -0.0479]$ and negative summer slopes in 31 of 33 years.

The calendar-matched calculation compares each high day with the mean of middle days no more than three or five calendar days away in the same month-year record. Eligible high-day contrasts are averaged within records before application of the original equal-month, equal-scale and equal-summer hierarchy. The ± 3 -day window matches 670 of 792 high days (84.6%); the ± 5 -day window matches 750 (94.7%). Each of the 99 evaluation month-year records contributes at least one match under both windows.

Table S12. Continuous-intensity and calendar-matched results for the 33 evaluation summers. Continuous entries are slopes of $\log Q_h$ per $^{\circ}\text{C}$. Matching entries are percentage effects. Brackets give 95% intervals.

Bandwidth	Continuous slope [interval]	± 3 -day match [interval]	± 5 -day match [interval]
126	-0.0318 [-0.0435, -0.0201]	-2.32 [-4.73, 0.09]	-2.54 [-4.90, -0.18]
252	-0.0408 [-0.0553, -0.0263]	-3.29 [-6.03, -0.56]	-3.54 [-6.18, -0.90]
503	-0.0546 [-0.0735, -0.0356]	-4.41 [-7.44, -1.38]	-4.74 [-7.66, -1.83]
1,006	-0.0877 [-0.1092, -0.0661]	-7.17 [-10.32, -4.02]	-7.90 [-10.94, -4.86]
2,013	-0.1104 [-0.1325, -0.0882]	-8.87 [-12.06, -5.68]	-9.89 [-12.98, -6.80]
Five-scale mean	-0.0650 [-0.0822, -0.0479]	-5.21 [-8.01, -2.41]	-5.72 [-8.43, -3.02]

The matched effects are weaker than the unrestricted high-to-middle contrast, but their broad-to-local ordering remains. Calendar matching also changes the target population of day pairs: only high days with eligible middle days enter the high-day average.

S10. Heavy-tailed ratio stress test

We compare the raw effect $r - 1$, the log effect $\log r$ and the bounded symmetric effect $2(r - 1)/(r + 1)$, where r is the high-to-middle dispersion ratio. The stress design uses the 121 observed site locations, exponential spatial covariance with 450-km range and 5% nugget, circular field correlation 0.45 and regional-mean autoregression 0.65. A common daily scale $\sqrt{3/\chi_3^2}$ produces heavy-tailed graph energy. Each scenario has 2,000 paired replications of six 30–31-day records. All three measures use the same 999 product shifts within a replication; the seed is 20260811.

Table S13. Raw, log and bounded effect measures under the heavy-tailed stress design. Bias and RMSE are on each measure's scale. Size and power are lower-tail rejection proportions at 0.05. Alternative targets correspond to a high-day energy multiplier of 0.93.

Measure	Null bias	Null RMSE	Size	Alternative target	Alt. bias	Alt. RMSE	Power
Raw ratio	0.3716	1.4725	0.0470	-0.0700	0.3456	1.3694	0.0795
Log ratio	-0.0698	0.3283	0.0490	-0.0726	-0.0698	0.3283	0.0850
Bounded symmetric	-0.0657	0.2771	0.0475	-0.0725	-0.0574	0.2748	0.0790

The three product-shift tests retain near-nominal size in this design. The raw ratio nevertheless has an RMSE more than five times that of the bounded measure under the null. Product-shift calibration concerns the randomisation distribution of a fixed statistic; ratio-estimation stability also depends on the right tail of daily energy. The log and bounded summaries reduce that instability, with the bounded measure giving the smallest RMSE in these cells.

S11. Simulation specifications and results

S11.1. Coupled label–field stress designs

The coupled simulation uses the 121 observed site coordinates and six records of lengths 30, 31, 30, 30, 31 and 30 days. Isotropic anomaly fields have exponential covariance with 450-km range and 5% nugget. The field state has autoregressive coefficient 0.45, and the regional weather series has coefficient 0.65. High and middle labels use type-7 quartiles within each record. A standardised spatial gradient is proportional to $0.8x + y$, where x and y are projected site coordinates. Its sign is driven by $0.75\tilde{\mu}_t + (1 - 0.75^2)^{1/2}Z_t$, so labels and spatial structure share a latent weather component even under the null.

Table S14. Exact changes defining the eight coupled label–field designs. Every cell retains the common six-record construction and within-record labels.

Design	Null	Contraction alternative
Shared latent gradient	Gradient amplitude is 1.8 on every day; its sign shares the latent driver with regional weather.	Gradient amplitude on high days is reduced from 1.8 to 1.2.
Within-month progression	Regional mean is a linear trend from -1.4 to 1.4 plus 0.60 times the AR(1) weather series; gradient amplitude remains 1.8.	Gradient amplitude declines linearly from 1.95 to 1.30 across the record.
Anisotropic covariance	Anomaly e-folding ranges are 800 km in projected longitude and 300 km in projected latitude.	High-day anomaly amplitude is multiplied by 0.94.
Peak-hour selection	Each daily regional mean is the maximum of 24 hourly values with sinusoidal amplitude 1.8 and independent noise SD 0.35; anomaly covariance is hour-invariant.	High-day anomaly amplitude is multiplied by 0.94.

Each field is evaluated with the five Gaussian graphs and a five-bin variogram profile whose bins are the quintiles of the 7,260 observed pair distances. For each replication, the target is the high-to-middle contrast in the conditional expected quadratic forms. The graph and variogram statistics use the same simulated fields, labels and product shifts. Each of the eight cells contains 1,000 replications with 499 shifts per test.

S11.2. Repeated-summer experiment

The annual experiment crosses $R \in \{20, 33, 60, 120\}$, $\rho \in \{0, 0.3, 0.6\}$, Gaussian, centred log-normal and standardised t_3 innovations, and effects $\delta \in \{0, -0.035, -0.07\}$. The resulting 108 cells each contain 10,000 replications after a 200-step burn-in. For retained summer r ,

$$U_r = \rho U_{r-1} + (1 - \rho^2)^{1/2} \varepsilon_r, \quad Y_r = \delta + 0.12 U_r.$$

The Gaussian innovation is standard normal. For $Z \sim N(0, 1)$, the centred unit-variance log-normal innovation is

$$\frac{\exp(0.65Z) - \exp(0.65^2/2)}{\{[\exp(0.65^2) - 1] \exp(0.65^2)\}^{1/2}},$$

and the heavy-tailed innovation is $T_3/\sqrt{3}$. The point estimate is $\bar{Y}$. The ordinary interval uses the sample variance and a t_{R-1} critical value. The lag-2 calculation uses a Bartlett long-run variance with the same critical value. The growing-lag interval uses $L = \max\{1, \lfloor R^{1/3} \rfloor\}$ and a standard-normal critical value. The sign test uses the exact binomial upper tail of the number of negative summers. The three-test decision requires the Student, lag-2 and sign p -values all to be at most 0.025. Standardised diagnostics use the known long-run variance $0.12^2(1 + \rho)/(1 - \rho)$.

The following tables report all eight coupled cells and all 108 annual cells.

Table S15. Complete results for the eight coupled label–field stress designs. Targets, bias and RMSE are percentage points; rejection is per cent over 1,000 replications. Graph and five-bin variogram profiles use identical fields, labels and product shifts.

Scenario	Family	Graph profile				Five-bin variogram			
		Target	Bias	RMSE	Reject	Target	Bias	RMSE	Reject
Shared-latent null	Null	0.00	0.38	3.63	5.5	-0.00	0.53	4.58	5.3
Shared-latent contraction	Alternative	-27.37	0.19	2.70	100.0	-36.66	0.25	3.26	100.0
Seasonal-progression null	Null	0.00	0.64	3.76	4.0	-0.00	0.85	4.77	4.3
Seasonal-gradient contraction	Alternative	-10.74	0.27	3.65	78.6	-14.72	0.33	4.60	83.3
Anisotropic null	Null	0.00	0.31	3.86	7.0	0.00	0.42	4.81	7.3
Anisotropic amplitude contraction	Alternative	-5.92	0.21	3.63	51.4	-3.98	0.32	4.58	25.5
Peak-selection null	Null	0.00	0.21	3.69	6.6	-0.00	0.34	4.63	6.6
Peak-selection contraction	Alternative	-5.90	0.36	3.50	54.2	-3.96	0.52	4.46	25.1

Table S16: Complete 108-cell repeated-summer results: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

R	ρ	Innovation	Effect	Bias percentage points	RMSE points	Student	Lag 2 coverage (%)	Growing
20	0.0	Gaussian	Null	0.015	2.655	95.4	91.7	89.9
20	0.0	Gaussian	-3.5%	-0.015	2.680	94.8	91.1	89.2
20	0.0	Gaussian	-7%	-0.015	2.665	95.1	91.3	89.4
20	0.0	Skewed	Null	-0.046	2.691	91.2	87.6	85.9
20	0.0	Skewed	-3.5%	0.022	2.676	92.1	88.7	86.9
20	0.0	Skewed	-7%	-0.024	2.662	91.8	88.3	86.5
20	0.0	t_3	Null	-0.028	2.672	95.3	91.7	89.7
20	0.0	t_3	-3.5%	0.028	2.599	95.9	92.0	90.4
20	0.0	t_3	-7%	0.028	2.696	95.9	92.0	90.1
20	0.3	Gaussian	Null	0.081	3.584	85.5	86.8	84.8
20	0.3	Gaussian	-3.5%	-0.096	3.597	85.0	86.6	84.2
20	0.3	Gaussian	-7%	0.036	3.620	85.0	86.5	84.0
20	0.3	Skewed	Null	0.053	3.612	82.7	84.4	82.3
20	0.3	Skewed	-3.5%	0.019	3.615	82.6	84.2	82.2
20	0.3	Skewed	-7%	-0.038	3.545	83.0	84.6	82.3
20	0.3	t_3	Null	0.056	3.486	85.9	87.5	85.1
20	0.3	t_3	-3.5%	-0.126	3.533	85.7	87.3	84.8
20	0.3	t_3	-7%	-0.002	3.620	85.8	87.8	85.3
20	0.6	Gaussian	Null	-0.042	5.096	66.5	76.2	73.4
20	0.6	Gaussian	-3.5%	0.002	5.128	67.0	76.3	73.8
20	0.6	Gaussian	-7%	0.000	5.175	66.6	75.9	73.2
20	0.6	Skewed	Null	0.038	5.187	64.4	73.9	71.2
20	0.6	Skewed	-3.5%	-0.043	5.140	64.5	74.3	71.5
20	0.6	Skewed	-7%	-0.039	5.115	65.1	74.6	71.7
20	0.6	t_3	Null	0.047	4.987	65.9	76.3	73.5
20	0.6	t_3	-3.5%	0.075	5.101	65.1	75.8	72.7
20	0.6	t_3	-7%	-0.026	5.254	65.5	76.7	73.7
33	0.0	Gaussian	Null	0.057	2.065	95.2	93.3	91.2
33	0.0	Gaussian	-3.5%	-0.004	2.078	95.2	93.1	91.2
33	0.0	Gaussian	-7%	-0.021	2.062	95.2	93.5	91.4
33	0.0	Skewed	Null	-0.029	2.071	92.4	90.6	88.5
33	0.0	Skewed	-3.5%	-0.002	2.061	92.9	91.0	89.0
33	0.0	Skewed	-7%	0.005	2.080	92.6	90.7	88.8
33	0.0	t_3	Null	-0.019	2.089	95.8	93.7	91.5
33	0.0	t_3	-3.5%	0.007	2.062	95.6	93.2	91.0
33	0.0	t_3	-7%	-0.004	2.127	95.7	93.6	91.5
33	0.3	Gaussian	Null	-0.046	2.837	85.4	89.0	87.3

Continued on next page

Table S16 continued

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
33	0.3	Gaussian	-3.5%	-0.002	2.819	85.4	88.6	87.2
33	0.3	Gaussian	-7%	0.018	2.808	85.7	88.8	87.3
33	0.3	Skewed	Null	0.000	2.795	83.3	87.2	85.9
33	0.3	Skewed	-3.5%	-0.004	2.834	82.9	86.4	85.1
33	0.3	Skewed	-7%	-0.008	2.792	83.4	86.8	85.4
33	0.3	t_3	Null	-0.011	2.773	85.0	89.2	87.5
33	0.3	t_3	-3.5%	0.012	2.804	85.5	89.4	88.2
33	0.3	t_3	-7%	0.033	2.760	85.4	89.1	87.8
33	0.6	Gaussian	Null	0.004	4.112	66.1	78.7	79.0
33	0.6	Gaussian	-3.5%	0.031	4.059	66.8	79.1	79.2
33	0.6	Gaussian	-7%	0.027	4.049	67.3	79.5	79.6
33	0.6	Skewed	Null	0.020	4.090	64.9	77.6	77.9
33	0.6	Skewed	-3.5%	0.043	4.053	65.6	77.8	78.0
33	0.6	Skewed	-7%	0.037	4.070	65.0	77.7	77.7
33	0.6	t_3	Null	-0.014	3.994	66.5	79.8	80.1
33	0.6	t_3	-3.5%	-0.054	4.004	66.0	79.9	80.3
33	0.6	t_3	-7%	0.006	4.158	65.4	79.6	79.8
60	0.0	Gaussian	Null	0.002	1.554	95.1	94.0	92.9
60	0.0	Gaussian	-3.5%	-0.018	1.562	94.8	93.6	92.4
60	0.0	Gaussian	-7%	-0.003	1.551	95.0	93.8	92.8
60	0.0	Skewed	Null	-0.002	1.553	93.4	92.2	91.1
60	0.0	Skewed	-3.5%	0.020	1.541	93.5	92.5	91.4
60	0.0	Skewed	-7%	0.039	1.545	94.1	93.2	92.0
60	0.0	t_3	Null	0.019	1.583	95.4	94.3	93.2
60	0.0	t_3	-3.5%	0.008	1.523	95.5	94.3	93.3
60	0.0	t_3	-7%	-0.008	1.536	95.7	94.8	93.6
60	0.3	Gaussian	Null	0.016	2.097	85.0	90.0	89.7
60	0.3	Gaussian	-3.5%	-0.016	2.068	85.9	90.9	90.4
60	0.3	Gaussian	-7%	0.033	2.083	85.6	90.6	90.3
60	0.3	Skewed	Null	-0.019	2.085	84.2	89.0	88.8
60	0.3	Skewed	-3.5%	0.018	2.096	84.1	89.1	88.8
60	0.3	Skewed	-7%	-0.001	2.097	83.7	88.6	88.4
60	0.3	t_3	Null	-0.014	2.082	85.2	90.5	90.0
60	0.3	t_3	-3.5%	0.035	2.088	84.7	90.7	90.3
60	0.3	t_3	-7%	-0.010	2.087	85.1	90.5	90.1
60	0.6	Gaussian	Null	-0.011	3.056	67.0	81.0	82.5
60	0.6	Gaussian	-3.5%	-0.029	3.066	66.9	81.5	83.0
60	0.6	Gaussian	-7%	0.030	3.036	67.3	81.3	82.8
60	0.6	Skewed	Null	0.009	3.026	65.9	80.1	81.5
60	0.6	Skewed	-3.5%	-0.016	3.062	65.4	79.9	81.5
60	0.6	Skewed	-7%	-0.018	3.038	66.5	80.2	81.6
60	0.6	t_3	Null	-0.003	2.988	66.1	81.0	82.6
60	0.6	t_3	-3.5%	0.005	3.063	66.6	81.3	82.9
60	0.6	t_3	-7%	0.026	3.046	66.0	81.5	83.0
120	0.0	Gaussian	Null	0.013	1.098	94.8	94.3	93.7
120	0.0	Gaussian	-3.5%	0.003	1.100	94.9	94.4	93.7
120	0.0	Gaussian	-7%	0.023	1.101	94.7	94.2	93.2
120	0.0	Skewed	Null	0.011	1.091	94.4	93.7	93.0
120	0.0	Skewed	-3.5%	0.003	1.084	94.2	93.5	93.1
120	0.0	Skewed	-7%	-0.004	1.101	93.7	93.3	92.5
120	0.0	t_3	Null	0.008	1.115	95.2	94.5	93.9
120	0.0	t_3	-3.5%	-0.007	1.099	95.2	94.4	93.8
120	0.0	t_3	-7%	0.004	1.100	95.5	94.8	94.0
120	0.3	Gaussian	Null	0.013	1.493	84.8	90.8	91.5
120	0.3	Gaussian	-3.5%	0.001	1.471	85.6	91.2	91.8
120	0.3	Gaussian	-7%	0.002	1.489	84.8	90.9	91.5
120	0.3	Skewed	Null	-0.004	1.502	83.6	89.6	90.2
120	0.3	Skewed	-3.5%	-0.014	1.485	84.7	90.2	90.7
120	0.3	Skewed	-7%	0.025	1.491	84.2	90.0	90.4
120	0.3	t_3	Null	-0.009	1.495	84.7	91.1	91.7

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Table S16 continued

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
120	0.3	t_3	-3.5%	-0.028	1.475	84.6	91.2	91.7
120	0.3	t_3	-7%	-0.002	1.488	84.5	90.7	91.4
120	0.6	Gaussian	Null	-0.018	2.164	67.9	82.8	86.5
120	0.6	Gaussian	-3.5%	-0.010	2.171	67.8	82.7	86.6
120	0.6	Gaussian	-7%	0.023	2.169	66.9	82.3	86.5
120	0.6	Skewed	Null	-0.012	2.162	67.2	81.8	85.6
120	0.6	Skewed	-3.5%	-0.020	2.181	65.7	81.3	85.6
120	0.6	Skewed	-7%	0.014	2.180	66.5	81.6	85.5
120	0.6	t_3	Null	-0.013	2.184	66.6	82.5	86.4
120	0.6	t_3	-3.5%	-0.053	2.146	66.5	82.5	86.6
120	0.6	t_3	-7%	0.015	2.180	65.7	82.2	86.3

Table S17: Complete 108-cell repeated-summer results: standardised-mean diagnostics. L is the growing Bartlett lag; the standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}, z_{.975}$
20	0.0	Gaussian	Null	2	0.006	0.990	-1.94, 1.98
20	0.0	Gaussian	-3.5%	2	-0.006	0.999	-1.98, 1.98
20	0.0	Gaussian	-7%	2	-0.006	0.993	-1.92, 1.99
20	0.0	Skewed	Null	2	-0.017	1.003	-1.71, 2.21
20	0.0	Skewed	-3.5%	2	0.008	0.997	-1.71, 2.18
20	0.0	Skewed	-7%	2	-0.009	0.992	-1.68, 2.14
20	0.0	t_3	Null	2	-0.010	0.996	-1.92, 1.95
20	0.0	t_3	-3.5%	2	0.011	0.968	-1.90, 1.90
20	0.0	t_3	-7%	2	0.010	1.005	-1.88, 1.91
20	0.3	Gaussian	Null	2	0.022	0.980	-1.91, 1.94
20	0.3	Gaussian	-3.5%	2	-0.026	0.983	-1.96, 1.93
20	0.3	Gaussian	-7%	2	0.010	0.990	-1.93, 1.92
20	0.3	Skewed	Null	2	0.014	0.988	-1.66, 2.18
20	0.3	Skewed	-3.5%	2	0.005	0.989	-1.68, 2.17
20	0.3	Skewed	-7%	2	-0.010	0.969	-1.68, 2.12
20	0.3	t_3	Null	2	0.015	0.953	-1.85, 1.91
20	0.3	t_3	-3.5%	2	-0.035	0.966	-1.94, 1.85
20	0.3	t_3	-7%	2	-0.001	0.990	-1.92, 1.91
20	0.6	Gaussian	Null	2	-0.008	0.950	-1.89, 1.83
20	0.6	Gaussian	-3.5%	2	0.000	0.956	-1.89, 1.85
20	0.6	Gaussian	-7%	2	0.000	0.964	-1.91, 1.88
20	0.6	Skewed	Null	2	0.007	0.967	-1.64, 2.10
20	0.6	Skewed	-3.5%	2	-0.008	0.958	-1.62, 2.11
20	0.6	Skewed	-7%	2	-0.007	0.953	-1.60, 2.11
20	0.6	t_3	Null	2	0.009	0.929	-1.77, 1.81
20	0.6	t_3	-3.5%	2	0.014	0.950	-1.81, 1.87
20	0.6	t_3	-7%	2	-0.005	0.979	-1.84, 1.79
33	0.0	Gaussian	Null	3	0.027	0.988	-1.92, 1.97
33	0.0	Gaussian	-3.5%	3	-0.002	0.995	-1.95, 1.92
33	0.0	Gaussian	-7%	3	-0.010	0.987	-1.93, 1.93
33	0.0	Skewed	Null	3	-0.014	0.991	-1.75, 2.12
33	0.0	Skewed	-3.5%	3	-0.001	0.987	-1.72, 2.11
33	0.0	Skewed	-7%	3	0.002	0.996	-1.73, 2.13
33	0.0	t_3	Null	3	-0.009	1.000	-1.93, 1.92
33	0.0	t_3	-3.5%	3	0.003	0.987	-1.93, 1.93
33	0.0	t_3	-7%	3	-0.002	1.018	-1.94, 1.91
33	0.3	Gaussian	Null	3	-0.016	0.996	-1.99, 1.95
33	0.3	Gaussian	-3.5%	3	-0.001	0.990	-1.93, 1.94
33	0.3	Gaussian	-7%	3	0.006	0.987	-1.92, 1.92
33	0.3	Skewed	Null	3	0.000	0.982	-1.72, 2.11
33	0.3	Skewed	-3.5%	3	-0.001	0.995	-1.71, 2.13

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Table S17 continued

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}, z_{.975}$
33	0.3	Skewed	-7%	3	-0.003	0.981	-1.73, 2.09
33	0.3	t_3	Null	3	-0.004	0.974	-1.93, 1.89
33	0.3	t_3	-3.5%	3	0.004	0.985	-1.92, 1.94
33	0.3	t_3	-7%	3	0.012	0.970	-1.94, 1.91
33	0.6	Gaussian	Null	3	0.001	0.984	-1.92, 1.92
33	0.6	Gaussian	-3.5%	3	0.008	0.972	-1.89, 1.87
33	0.6	Gaussian	-7%	3	0.006	0.969	-1.90, 1.88
33	0.6	Skewed	Null	3	0.005	0.979	-1.70, 2.12
33	0.6	Skewed	-3.5%	3	0.010	0.970	-1.68, 2.10
33	0.6	Skewed	-7%	3	0.009	0.974	-1.68, 2.09
33	0.6	t_3	Null	3	-0.003	0.956	-1.87, 1.83
33	0.6	t_3	-3.5%	3	-0.013	0.958	-1.89, 1.82
33	0.6	t_3	-7%	3	0.001	0.995	-1.96, 1.88
60	0.0	Gaussian	Null	3	0.001	1.003	-1.93, 1.95
60	0.0	Gaussian	-3.5%	3	-0.012	1.008	-2.01, 1.94
60	0.0	Gaussian	-7%	3	-0.002	1.001	-1.94, 1.98
60	0.0	Skewed	Null	3	-0.001	1.003	-1.79, 2.10
60	0.0	Skewed	-3.5%	3	0.013	0.995	-1.79, 2.09
60	0.0	Skewed	-7%	3	0.025	0.997	-1.78, 2.15
60	0.0	t_3	Null	3	0.013	1.022	-1.96, 2.01
60	0.0	t_3	-3.5%	3	0.005	0.983	-1.92, 1.96
60	0.0	t_3	-7%	3	-0.005	0.991	-1.89, 1.92
60	0.3	Gaussian	Null	3	0.008	0.993	-1.99, 1.92
60	0.3	Gaussian	-3.5%	3	-0.007	0.979	-1.92, 1.90
60	0.3	Gaussian	-7%	3	0.015	0.986	-1.91, 1.91
60	0.3	Skewed	Null	3	-0.009	0.988	-1.81, 2.07
60	0.3	Skewed	-3.5%	3	0.008	0.993	-1.78, 2.13
60	0.3	Skewed	-7%	3	-0.001	0.993	-1.83, 2.08
60	0.3	t_3	Null	3	-0.007	0.986	-1.92, 1.98
60	0.3	t_3	-3.5%	3	0.017	0.989	-1.91, 1.96
60	0.3	t_3	-7%	3	-0.005	0.988	-1.96, 1.95
60	0.6	Gaussian	Null	3	-0.003	0.986	-1.92, 1.94
60	0.6	Gaussian	-3.5%	3	-0.009	0.989	-1.96, 1.90
60	0.6	Gaussian	-7%	3	0.010	0.980	-1.89, 1.92
60	0.6	Skewed	Null	3	0.003	0.977	-1.78, 2.06
60	0.6	Skewed	-3.5%	3	-0.005	0.988	-1.77, 2.10
60	0.6	Skewed	-7%	3	-0.006	0.980	-1.76, 2.05
60	0.6	t_3	Null	3	-0.001	0.964	-1.90, 1.87
60	0.6	t_3	-3.5%	3	0.002	0.989	-1.89, 1.92
60	0.6	t_3	-7%	3	0.008	0.983	-1.94, 1.92
120	0.0	Gaussian	Null	4	0.012	1.003	-1.98, 1.97
120	0.0	Gaussian	-3.5%	4	0.002	1.004	-1.93, 1.99
120	0.0	Gaussian	-7%	4	0.021	1.005	-1.94, 2.02
120	0.0	Skewed	Null	4	0.010	0.996	-1.82, 2.07
120	0.0	Skewed	-3.5%	4	0.003	0.989	-1.86, 2.05
120	0.0	Skewed	-7%	4	-0.003	1.005	-1.88, 2.07
120	0.0	t_3	Null	4	0.007	1.018	-1.95, 2.01
120	0.0	t_3	-3.5%	4	-0.006	1.003	-1.96, 1.96
120	0.0	t_3	-7%	4	0.004	1.004	-1.92, 2.02
120	0.3	Gaussian	Null	4	0.008	1.000	-1.98, 1.95
120	0.3	Gaussian	-3.5%	4	0.001	0.986	-1.92, 1.94
120	0.3	Gaussian	-7%	4	0.001	0.997	-1.94, 1.95
120	0.3	Skewed	Null	4	-0.003	1.006	-1.86, 2.10
120	0.3	Skewed	-3.5%	4	-0.009	0.995	-1.87, 2.06
120	0.3	Skewed	-7%	4	0.017	0.999	-1.82, 2.06
120	0.3	t_3	Null	4	-0.006	1.001	-1.94, 1.93
120	0.3	t_3	-3.5%	4	-0.018	0.988	-1.95, 1.94
120	0.3	t_3	-7%	4	-0.001	0.997	-1.98, 1.94
120	0.6	Gaussian	Null	4	-0.008	0.988	-1.97, 1.95
120	0.6	Gaussian	-3.5%	4	-0.005	0.991	-1.97, 1.94

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Table S17 continued

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}, z_{.975}$
120	0.6	Gaussian	-7%	4	0.010	0.990	-1.92, 1.93
120	0.6	Skewed	Null	4	-0.005	0.987	-1.83, 2.09
120	0.6	Skewed	-3.5%	4	-0.009	0.996	-1.85, 2.06
120	0.6	Skewed	-7%	4	0.006	0.995	-1.82, 2.08
120	0.6	t_3	Null	4	-0.006	0.997	-1.87, 1.91
120	0.6	t_3	-3.5%	4	-0.024	0.979	-1.97, 1.90
120	0.6	t_3	-7%	4	0.007	0.995	-1.95, 1.93

Table S18: Complete 108-cell repeated-summer results: one-sided rejection percentages. The three-test column requires Student, lag-2 and sign values all to be at most 0.025. Under non-null cells the entries are power, not size.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
20	0.0	Gaussian	Null	4.6	7.1	8.0	2.1	1.0
20	0.0	Gaussian	-3.5%	34.8	40.7	43.6	15.8	12.1
20	0.0	Gaussian	-7%	81.0	84.3	86.2	49.5	46.4
20	0.0	Skewed	Null	11.8	14.6	15.6	19.0	6.3
20	0.0	Skewed	-3.5%	45.4	49.5	51.6	53.8	31.2
20	0.0	Skewed	-7%	77.3	80.3	81.5	82.9	64.7
20	0.0	t_3	Null	5.2	7.5	8.4	2.2	0.9
20	0.0	t_3	-3.5%	43.9	49.8	52.3	32.7	21.2
20	0.0	t_3	-7%	85.7	87.7	88.8	82.8	71.0
20	0.3	Gaussian	Null	10.5	9.3	10.4	4.8	3.1
20	0.3	Gaussian	-3.5%	41.0	37.6	39.9	21.9	17.9
20	0.3	Gaussian	-7%	74.5	70.5	72.6	51.0	46.9
20	0.3	Skewed	Null	17.4	16.5	17.6	20.3	10.3
20	0.3	Skewed	-3.5%	47.8	44.6	46.3	50.9	33.3
20	0.3	Skewed	-7%	75.2	72.7	74.1	77.2	62.1
20	0.3	t_3	Null	11.3	10.0	11.0	5.5	3.1
20	0.3	t_3	-3.5%	48.8	44.7	46.9	35.2	25.9
20	0.3	t_3	-7%	81.3	78.0	79.7	75.4	64.6
20	0.6	Gaussian	Null	20.8	15.9	17.1	11.8	9.3
20	0.6	Gaussian	-3.5%	44.4	36.0	37.7	28.7	24.2
20	0.6	Gaussian	-7%	70.2	62.0	63.8	52.5	47.9
20	0.6	Skewed	Null	27.0	22.1	23.0	24.1	16.9
20	0.6	Skewed	-3.5%	51.7	44.9	46.5	47.2	36.5
20	0.6	Skewed	-7%	71.8	65.3	66.7	68.5	57.1
20	0.6	t_3	Null	21.5	16.0	17.2	12.6	9.4
20	0.6	t_3	-3.5%	49.2	40.8	42.4	36.9	29.6
20	0.6	t_3	-7%	76.2	68.8	70.4	66.9	58.6
33	0.0	Gaussian	Null	4.8	6.0	7.1	3.6	0.8
33	0.0	Gaussian	-3.5%	49.6	53.2	56.3	33.8	18.1
33	0.0	Gaussian	-7%	95.0	95.6	96.1	81.4	68.5
33	0.0	Skewed	Null	10.3	12.0	13.3	39.3	5.9
33	0.0	Skewed	-3.5%	55.5	58.1	60.6	84.7	42.7
33	0.0	Skewed	-7%	89.0	89.8	90.8	98.4	81.8
33	0.0	t_3	Null	5.0	6.5	7.8	4.0	0.7
33	0.0	t_3	-3.5%	58.2	61.0	63.6	62.0	33.2
33	0.0	t_3	-7%	94.4	94.9	95.6	98.3	87.9
33	0.3	Gaussian	Null	11.8	9.1	9.9	8.2	3.0
33	0.3	Gaussian	-3.5%	50.5	44.1	45.3	37.0	23.0
33	0.3	Gaussian	-7%	89.2	85.1	85.5	77.5	63.8
33	0.3	Skewed	Null	17.1	14.4	14.9	36.5	9.4
33	0.3	Skewed	-3.5%	55.7	50.7	51.6	76.8	40.7
33	0.3	Skewed	-7%	85.2	81.8	82.3	95.2	74.7
33	0.3	t_3	Null	11.7	9.3	10.0	9.1	3.3
33	0.3	t_3	-3.5%	57.8	51.2	52.2	57.5	33.7
33	0.3	t_3	-7%	90.7	87.9	88.2	94.3	79.5

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Table S18 continued

<i>R</i>	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
33	0.6	Gaussian	Null	21.2	14.4	14.1	15.9	8.5
33	0.6	Gaussian	−3.5%	50.8	40.2	39.4	41.1	27.8
33	0.6	Gaussian	−7%	80.7	72.4	71.4	71.5	59.2
33	0.6	Skewed	Null	26.1	20.0	19.5	34.3	14.9
33	0.6	Skewed	−3.5%	55.5	46.7	45.9	65.2	39.0
33	0.6	Skewed	−7%	79.5	72.0	71.2	85.6	65.1
33	0.6	t_3	Null	21.6	14.4	14.1	16.5	8.1
33	0.6	t_3	−3.5%	56.9	46.7	45.8	52.6	34.8
33	0.6	t_3	−7%	84.8	77.6	76.8	84.6	68.4
60	0.0	Gaussian	Null	5.0	5.8	6.5	4.9	0.8
60	0.0	Gaussian	−3.5%	71.8	73.6	74.7	54.4	31.7
60	0.0	Gaussian	−7%	99.7	99.7	99.7	96.8	90.7
60	0.0	Skewed	Null	9.2	10.0	10.6	62.6	5.2
60	0.0	Skewed	−3.5%	70.6	71.7	72.7	98.3	59.5
60	0.0	Skewed	−7%	97.6	97.7	97.9	100.0	95.5
60	0.0	t_3	Null	4.7	5.5	6.2	4.7	0.5
60	0.0	t_3	−3.5%	77.5	78.2	79.1	87.0	57.0
60	0.0	t_3	−7%	98.8	98.7	98.8	100.0	97.6
60	0.3	Gaussian	Null	11.2	8.2	8.3	8.9	2.8
60	0.3	Gaussian	−3.5%	67.5	59.5	59.4	54.2	33.7
60	0.3	Gaussian	−7%	98.4	97.0	96.9	93.9	86.0
60	0.3	Skewed	Null	15.2	11.9	12.1	52.5	8.3
60	0.3	Skewed	−3.5%	67.5	61.3	61.1	93.7	52.2
60	0.3	Skewed	−7%	95.5	93.2	93.1	99.7	89.6
60	0.3	t_3	Null	11.8	8.1	8.1	10.0	2.4
60	0.3	t_3	−3.5%	71.9	64.7	64.5	77.5	47.9
60	0.3	t_3	−7%	97.4	96.5	96.3	99.6	93.9
60	0.6	Gaussian	Null	21.2	13.1	12.3	16.8	7.3
60	0.6	Gaussian	−3.5%	63.3	50.9	48.8	53.3	35.7
60	0.6	Gaussian	−7%	92.7	87.0	85.7	87.2	75.9
60	0.6	Skewed	Null	24.4	17.9	16.9	42.5	13.4
60	0.6	Skewed	−3.5%	64.4	54.2	52.3	80.7	46.7
60	0.6	Skewed	−7%	90.1	84.3	83.2	96.5	79.1
60	0.6	t_3	Null	21.6	13.6	12.5	17.7	7.4
60	0.6	t_3	−3.5%	66.9	56.5	54.2	66.6	43.3
60	0.6	t_3	−7%	92.9	88.4	87.5	95.5	83.1
120	0.0	Gaussian	Null	5.1	5.5	5.9	4.0	0.7
120	0.0	Gaussian	−3.5%	93.3	93.6	93.8	78.8	65.7
120	0.0	Gaussian	−7%	100.0	100.0	100.0	100.0	99.8
120	0.0	Skewed	Null	7.8	8.0	8.4	86.3	4.2
120	0.0	Skewed	−3.5%	90.3	90.5	90.8	100.0	83.8
120	0.0	Skewed	−7%	100.0	99.9	99.9	100.0	99.9
120	0.0	t_3	Null	4.9	5.2	5.6	4.2	0.6
120	0.0	t_3	−3.5%	93.5	93.6	93.9	98.8	87.1
120	0.0	t_3	−7%	99.7	99.8	99.8	100.0	99.6
120	0.3	Gaussian	Null	11.0	7.9	7.4	7.9	2.9
120	0.3	Gaussian	−3.5%	87.2	81.7	80.2	74.4	60.5
120	0.3	Gaussian	−7%	100.0	100.0	100.0	99.6	99.2
120	0.3	Skewed	Null	14.8	11.0	10.4	71.9	7.4
120	0.3	Skewed	−3.5%	84.6	79.2	77.9	99.7	71.6
120	0.3	Skewed	−7%	99.6	99.2	99.2	100.0	98.6
120	0.3	t_3	Null	12.0	7.9	7.4	8.8	2.3
120	0.3	t_3	−3.5%	88.8	84.5	83.4	94.2	75.5
120	0.3	t_3	−7%	99.7	99.4	99.4	100.0	99.1
120	0.6	Gaussian	Null	20.4	12.5	10.2	15.9	7.2
120	0.6	Gaussian	−3.5%	78.9	67.5	62.8	68.8	53.9
120	0.6	Gaussian	−7%	99.2	98.2	97.2	97.7	95.0
120	0.6	Skewed	Null	23.4	15.9	13.4	53.8	11.9
120	0.6	Skewed	−3.5%	77.5	67.9	63.2	94.3	60.0
120	0.6	Skewed	−7%	97.9	95.6	94.3	99.9	93.2

Continued on next page

Table S18 continued

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
120	0.6	t_3	Null	21.9	13.5	10.9	17.3	7.2
120	0.6	t_3	-3.5%	81.0	71.7	67.3	83.3	62.1
120	0.6	t_3	-7%	98.7	97.4	96.7	99.8	96.0

S11.3. Cross-record dependence stress test

Proposition 1 requires joint invariance under independent rotations of all monthly records. Marginal cyclic invariance of each record does not imply this product condition. The stress test uses the 121 observed site coordinates, the five Gaussian graphs and six records from two simulated summers. The June, July and August record lengths were 30, 31 and 31 days in each summer.

For summer y , let

$$U_y(u) = \sum_{k=1}^3 \sqrt{w_k} \{A_{yk} \cos(2\pi ku) + B_{yk} \sin(2\pi ku)\}, \quad u \in [0, 1),$$

where the coefficients are independent standard normal variables and $(w_1, w_2, w_3) = (1, 0.45, 0.20)/1.65$. The same coefficients were used for all three records in a summer. For month m with length n_m , the centred spatial field at $t = 0, \dots, n_m - 1$ was

$$\mathbf{z}_{ymt} = \lambda U_y(t/n_m) \mathbf{g} + \sqrt{1 - \lambda^2} \boldsymbol{\varepsilon}_{ymt}, \quad \lambda \in \{0, 0.3, 0.6, 0.8\}.$$

The vector $\mathbf{g}$ is the centred, unit-mean-square pattern proportional to $0.8x + y$ in projected coordinates. The month-specific Gaussian noise used the first 12 square-root-eigenvalue-weighted eigenmodes of the centred spatial covariance $0.95 \exp(-d/450) + 0.05I$, where d is in km. The retained basis was rescaled to mean site variance one. Its circular moving-average filter had weights $(0.20, 0.50, 1, 0.50, 0.20)$, normalised to unit Euclidean norm. Regional-mean series X_{ymt} were generated from an independent random-number stream with a circular filter proportional to $(0.25, 0.60, 1, 0.60, 0.25)$. The simulated field is $\mathbf{y}_{ymt} = X_{ymt} \mathbf{1} + \mathbf{z}_{ymt}$; within-record type-7 quartiles of X_{ymt} define the high and middle labels.

Each individual record has a cyclically stationary Gaussian distribution. The shared Fourier coefficients align the three monthly phases within a summer, so independent month-wise rotations alter their cross-covariance when $\lambda > 0$. A numerical covariance check gave a maximum marginal one-step rotation error of 2.0×10^{-15} . Independently rotating the 30-day record by one day changed its cross-covariance with a 31-day record by 0.275 in relative Frobenius norm; the corresponding graph-dispersion cross-covariance changed by 0.362. The pooled same-summer July–August correlation of corresponding daily mean-dispersion values, averaged over the five graphs, increased from 0.003 at $\lambda = 0$ to 0.769 at $\lambda = 0.8$.

For each value of λ , we generated 2,000 replications and used 999 independent product shifts per replication. The statistic averaged the record-specific high-to-middle relative contrasts (ratios minus one) equally over the six records and five graphs. Table S19 reports lower-tail rejection at $\alpha = 0.05$; the seed was 20260812.

Table S19. Product-shift size under shared cross-record temporal dependence.

Shared loading λ	Rejection rate	Monte Carlo SE
0.0	0.0515	0.00494
0.3	0.0510	0.00492
0.6	0.0495	0.00485
0.8	0.0490	0.00483

Rejection remained between 4.90% and 5.15% as the shared component increased. Thus rejection remained near its nominal level for these simulated departures from joint product invariance.

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